

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study analyzed aggregated, non-personal PhilGEPS medical procurement records using unsupervised machine learning to identify procurement behavior patterns and potential systemic risk indicators, including delays, high-value or high-volume procurement, supplier concentration, and inefficiencies.

Following CRISP-DM, the dataset (487,605 records from 2020 Q2 to 2025 Q3) was cleaned, filtered, transformed, and feature-engineered before applying scaling and PCA for dimensionality reduction and visualization.

K-means was used as the primary clustering method, supported by evaluation metrics (silhouette score, Davies-Bouldin index, Calinski-Harabasz index, composite scoring), resulting in the selection of $K = 6$ clusters representing distinct procurement behavior segments. DBSCAN was also applied for density-based clustering and outlier detection but served only as a comparative method due to sensitivity and lower interpretability.

The resulting clusters were interpreted into meaningful procurement profiles using PCA plots and feature-based analysis. A web-based MedFlow PH dashboard was developed using React, Tailwind CSS, and Node.js to present preprocessing results, visualizations, clustering outputs, and evaluations.



Overall, the study demonstrates that unsupervised clustering can reveal meaningful procurement patterns in large-scale public health procurement data, providing decision-support insights for monitoring and resource allocation without directly implying actual shortages or oversupply.

Conclusions

Following the accomplishment of the research objectives and the evaluation of the clustering outputs and dashboard presentation, the following conclusions were drawn.

Conclusion for Objective 1: Analysis of Aggregated Medical-Related Procurement Records from PhilGEPS

1. The study concluded that aggregated PhilGEPS medical procurement data contains sufficient variation to support unsupervised clustering and reveal broad procurement behavior patterns across budgets, quantities, timing, regions, and agencies. From 487,605 records, the analysis highlighted the importance of thorough data cleaning, deduplication, filtering, and transformation to address issues such as missing values, inconsistent formats, and schema differences before applying machine learning. Overall,



the objective was achieved by demonstrating that properly prepared procurement data can be effectively used for pattern discovery and further analytical modeling.

Conclusion for Objective 2: Application of K-means and DBSCAN Clustering Algorithms

2. The study concluded that unsupervised clustering can effectively group medical-related procurement records into meaningful behavior segments. K-means produced six well-defined and interpretable clusters, supported by multiple validation metrics, representing distinct procurement patterns such as high-value centralized procurement, routine supply procurement, and delayed procurement activities. DBSCAN provided additional insights by detecting density patterns and outliers, but was less stable due to parameter sensitivity and weaker interpretability. Overall, K-means was identified as the primary model for analysis, with DBSCAN serving as a complementary method for identifying noise and density-based structures.

Conclusion for Objective 3: Development of the MedFlow PH Web-Based Results Dashboard

3. The study concluded that the MedFlow PH web-based dashboard effectively presented the clustering results and analysis outputs in a clear



and structured format. Built using React, Tailwind CSS, and Node.js, it organized the full workflow—from data cleaning and preprocessing to clustering, evaluation, interpretation, and model comparison—through visualizations such as PCA plots, heatmaps, and performance metrics. The dashboard successfully translated complex analytical results into accessible insights for students, researchers, and public health stakeholders, achieving its objective of clear and effective results communication.

Conclusion for Objective 4: Evaluation and Comparison of K-means and DBSCAN Results

4. The study concluded that K-means was the more suitable primary clustering model for MedFlow PH due to its more stable, interpretable, and structured results compared to DBSCAN. Based on K-selection diagnostics, $K = 6$ was chosen, with a best silhouette score of 0.386, indicating moderate but acceptable cluster separation for real-world procurement data. Multiple evaluation metrics, including Davies-Bouldin Index, Calinski-Harabasz Index, and visual PCA inspection, supported this choice. While DBSCAN was useful for detecting noise and density-based patterns, it was less interpretable for defining clear procurement segments. Overall, K-means



better achieved the study's goal of producing understandable and meaningful procurement behavior clusters.

Recommendations

Based on the conclusions drawn from the study, the following recommendations are proposed.

Recommendation for Objective 1

1. Future studies should improve the quality and completeness of PhilGEPS medical procurement data, addressing missing values, duplicates, inconsistent formats, and schema differences through systematic cleaning and validation. Incorporating additional data sources—such as inventory, delivery, consumption, and demand records would strengthen cluster interpretation and reduce reliance on procurement data alone. Using more detailed but privacy-safe datasets with contextual variables like facility type, regional demand, lead times, supplier performance, and delivery status could further enhance analysis.



Recommendation for Objective 2

2. Future studies should explore additional clustering algorithms and feature sets, test wider K ranges for K-means, refine DBSCAN parameters or use HDBSCAN, and interpret clusters cautiously as analytical indicators rather than direct proof of procurement issues.

Recommendation for Objective 3

3. Future development of the MedFlow PH dashboard should focus on interactivity and user-friendliness, including dynamic filtering by year, region, agency, procurement category, cluster label, or amount. Enhancing explanatory notes, adding downloadable reports, interactive tables, and export options will improve usability for stakeholders. Maintaining a modular design will simplify updates and enable convenient online access to new results and visualizations.

Recommendation for Objective 4

4. Future studies should evaluate clustering models using both numerical metrics (Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index) and interpretability to ensure practical usefulness. Cluster labels should be validated against external indicators like stockouts, delivery delays,



emergency procurement, supplier performance, and facility demand. Stakeholder review by public health administrators, procurement officers, and supply chain managers is recommended, along with additional model comparisons such as cluster stability, outlier detection, runtime, and sensitivity analysis.

Additional Recommendations

Data Governance and Documentation

Establish clear documentation of data sources, cleaning rules, transformations, and model parameters to ensure transparency, reproducibility, and easier future improvements.

Ethical and Privacy Considerations

Maintain strict data privacy practices, including anonymization and access control, especially if more detailed datasets are introduced in future studies.

Future Research Expansion

Expand analysis across regions, facilities, time periods, procurement types, and suppliers, and include time-series analysis to track changes over time.



Final Statement

The study demonstrates that unsupervised learning can identify procurement patterns in PhilGEPS data, providing useful decision-support insights for public health procurement. While it does not confirm actual shortages or inefficiencies, the results can guide further investigation, monitoring, and resource planning, and serve as a foundation for more advanced procurement analytics.